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1. Create a VIEW that displays all buses starting from a source name starting with “ISBT".

```
mysql> CREATE VIEW ISBT_BUSES AS
    -> SELECT *
    -> FROM LIST_BUSES
    -> WHERE Source LIKE 'ISBT%';
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM ISBT_BUSES;
+--------+------------+--------+-------------+------+------------+----------+
| Bus_no | Bus_name   | Source | Destination | Fare | Date       | Time     |
+--------+------------+--------+-------------+------+------------+----------+
| DL306  | Shatabdi   | ISBT   | Dehradun    | 1007 | 2022-09-29 | 20:45:00 |
| DL701  | Vaishnavi  | ISBT   | Chandigarh  | 1550 | 2022-09-28 | 23:55:00 |
| HR303  | Maheshwaram | ISBT   | Ambala      | 1010 | 2022-09-25 | 07:22:00 |
| UP505  | Goluxury   | ISBT   | Kanpur      | 2250 | 2022-09-27 | 19:38:00 |
+--------+------------+--------+-------------+------+------------+----------+
4 rows in set (0.01 sec)
```

2. Display source locations that have more than 2 buses originating from them.

```
mysql> SELECT Source, COUNT(Bus_no) AS Total_Buses
    -> FROM List_Buses
    -> GROUP BY Source
    -> HAVING COUNT(Bus_no) > 2;
+------------+-------------+
| Source     | Total_Buses |
+------------+-------------+
| ISBT       |           4 |
| Pari Chauk |           3 |
+------------+-------------+
2 rows in set (0.01 sec)
```

3. Show bus records that are scheduled to depart after September 25, 2022, ordered by departure date and time.

```
mysql> SELECT *
    -> FROM List_Buses
    -> WHERE Date > '2022-09-25'
    -> ORDER BY Date, Time;
+--------+-----------+------------+-------------+------+------------+----------+
| Bus_no | Bus_name  | Source     | Destination | Fare | Date       | Time     |
+--------+-----------+------------+-------------+------+------------+----------+
| UP505  | Goluxury  | ISBT       | Kanpur      | 2250 | 2022-09-27 | 19:38:00 |
| DL701  | Vaishnavi | ISBT       | Chandigarh  | 1550 | 2022-09-28 | 23:55:00 |
| DL306  | Shatabdi  | ISBT       | Dehradun    | 1007 | 2022-09-29 | 20:45:00 |
| UP501  | Safar     | Pari Chauk | Varanasi    | 1080 | 2022-09-30 | 08:35:00 |
+--------+-----------+------------+-------------+------+------------+----------+
4 rows in set (0.01 sec)
```

4. Create a view named "Confirmed_Passengers" that displays details of passengers with a confirmed booking status.

```
mysql> CREATE VIEW Confirmed_Passengers AS
    -> SELECT *
    -> FROM Passengers
    -> WHERE Status = 'Confirm';
Query OK, 0 rows affected (0.03 sec)

mysql> SELECT * FROM  Confirmed_Passengers;
+--------------+----------+--------+------+--------+---------+
| Passenger_id | P_name   | Bus_no | Age  | Gender | Status  |
+--------------+----------+--------+------+--------+---------+
| SCS012       | Arjun    | UP301  |   17 | M      | Confirm |
| SCS013       | Anamika  | HR302  |   15 | F      | Confirm |
| SCS014       | Diya     | HR303  |   19 | F      | Confirm |
| SCS015       | Abhishek | UP505  |   57 | M      | Confirm |
| SCS043       | Rupam    | UP501  |   22 | F      | Confirm |
| SCS056       | Satya    | UP101  |   23 | M      | Confirm |
| SCS071       | Rahul    | DL306  |   21 | M      | Confirm |
| SCS088       | ANKIT    | UP500  |   24 | M      | CONFIRM |
+--------------+----------+--------+------+--------+---------+
8 rows in set (0.00 sec)
```

5. Retrieve the details of confirmed passengers from the “Confirmed_Passengers" view, ordered by age in descending order, and then by passenger names.

```
mysql> SELECT *
    -> FROM Confirmed_Passengers
    -> ORDER BY Age DESC, P_Name;
+--------------+----------+--------+------+--------+---------+
| Passenger_id | P_name   | Bus_no | Age  | Gender | Status  |
+--------------+----------+--------+------+--------+---------+
| SCS015       | Abhishek | UP505  |   57 | M      | Confirm |
| SCS088       | ANKIT    | UP500  |   24 | M      | CONFIRM |
| SCS056       | Satya    | UP101  |   23 | M      | Confirm |
| SCS043       | Rupam    | UP501  |   22 | F      | Confirm |
| SCS071       | Rahul    | DL306  |   21 | M      | Confirm |
| SCS014       | Diya     | HR303  |   19 | F      | Confirm |
| SCS012       | Arjun    | UP301  |   17 | M      | Confirm |
| SCS013       | Anamika  | HR302  |   15 | F      | Confirm |
+--------------+----------+--------+------+--------+---------+
8 rows in set (0.01 sec)
```

6. Retrieve the total count of passengers in each status category, ordered by count in descending order.

```
mysql> SELECT Status, COUNT(*) AS Total_Passengers
    -> FROM Passengers
    -> GROUP BY Status
    -> ORDER BY Total_Passengers DESC;
+---------+------------------+
| Status  | Total_Passengers |
+---------+------------------+
| Confirm |                8 |
| Waiting |                4 |
+---------+------------------+
2 rows in set (0.00 sec)
```

7. Create a view that lists buses that are fully booked.

```
mysql> CREATE VIEW Fully_Booked_Buses AS
    -> SELECT *
    -> FROM Bus_Status
    -> WHERE Available_seats = 0;
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM Fully_Booked_Buses;
Empty set (0.00 sec)
```

8. Create a view that displays buses with low available seats (where available seats are less than 10).

```
mysql> CREATE VIEW LOW_Seats_Buses AS
    -> SELECT *
    -> FROM Bus_Status
    -> where Available_seats<10;
Query OK, 0 rows affected (0.02 sec)

mysql> select * from LOW_Seats_Buses;
+--------+----------------+--------------+
| Bus_no | Available_seats | Booked_seats |
+--------+----------------+--------------+
| UP505  |              1 |           59 |
| DL701  |              8 |           46 |
+--------+----------------+--------------+
2 rows in set (0.00 sec)
```

9. Retrieve a list of buses ordered by the number of booked seats in descending order.

```
mysql> SELECT *
    -> FROM Bus_Status
    -> ORDER BY Booked_seats DESC;
+--------+----------------+--------------+
| Bus_no | Available_seats | Booked_seats |
+--------+----------------+--------------+
| UP505  |              1 |           59 |
| UP101  |             10 |           51 |
| HR303  |             14 |           48 |
| DL701  |              8 |           46 |
| DL306  |             17 |           45 |
| UP301  |             35 |           25 |
| HR302  |             41 |           24 |
| UP501  |             50 |            0 |
+--------+----------------+--------------+
8 rows in set (0.00 sec)
```

10. Retrieve a list of buses ordered by the total number of seats (available seats + booked seats) in ascending order.

```
mysql> SELECT Bus_no,
    ->        (Available_seats + Booked_seats) AS Total_Seats
    -> FROM Bus_Status
    -> ORDER BY Total_Seats ASC;
+--------+-------------+
| Bus_no | Total_Seats |
+--------+-------------+
| UP501  |          50 |
| DL701  |          54 |
| UP301  |          60 |
| UP505  |          60 |
| UP101  |          61 |
| HR303  |          62 |
| DL306  |          62 |
| HR302  |          65 |
+--------+-------------+
8 rows in set (0.00 sec)
```